Alpha005 MacroReflexivity

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1 Philosophy: why reflexivity

This alpha is inspired by George Soros' reflexivity theory. The core idea is:

- prices affect beliefs,
- beliefs affect positioning,
- positions affect prices.

This generates a feedback loop. For example, in an upward trend:

$$\text{price rises} \implies \text{traders become more bullish} \implies \text{new capital pushes the price higher}$$

The reflexivity effect is stronger for assets that lack strong fundamental support, for example the crypto market or high-growth small companies.

This alpha does not try to estimate a "true value" of the asset. Instead, it treats the market as a dynamic system around a slow-moving equilibrium. The key question is not only whether price is above or below equilibrium, but also whether the deviation from equilibrium is expanding or shrinking. If the deviation keeps expanding in the same direction, the strategy interprets it as a self-reinforcing trend. If the deviation starts shrinking, the strategy interprets it as trend decay or reversion.

2 Maths Principle

We define the following notations:

- S_t = equilibrium; in practice, this is a slow moving average, and it acts like a reference level.
- F_t = short-term state of price; in practice, this is a fast moving average.
- D_t = normalised deviation
$$D_t = \frac{F_t - S_t}{S_t} \tag{1}$$
- V_t = change in deviation (discrete),
$$V_t = D_t - D_{t-1} \tag{2}$$
- R_t = change of deviation energy,
$$R_t = D_t \cdot V_t \tag{3}$$

2.1 What does D_t mean?

The sign of D_t indicates which side of equilibrium the market is on at this moment.

- if $D_t > 0$, the market is above equilibrium, which is interpreted as bullish.
- if $D_t < 0$, the market is below equilibrium, which is interpreted as bearish.

However, this is not enough, because being above equilibrium does not tell you whether price will move even further away or start moving back. Therefore, we introduce the change of deviation V_t.

2.2 What does V_t mean?

- if $V_t > 0$, the deviation is becoming more positive, and the market is shifting upward relative to equilibrium.
- if $V_t < 0$, the deviation is becoming more negative, and the market is shifting downward relative to equilibrium.

Remark. $V_t > 0$ does not always mean bullish strengthening; it depends on where you started. For example,

$$D_{t-1} = -0.08, \quad D_t = -0.04 \implies V_t = 0.04$$

In this case, the market is still below equilibrium, but it is recovering toward equilibrium rather than strengthening downward.

2.3 What does R_t mean?

The quantity R_t combines the current deviation and the change in deviation:

$$R_t = D_t(D_t - D_{t-1}) \tag{4}$$

It tells us whether the market is moving farther away from equilibrium or back toward it.

- if $R_t > 0$, the market is moving farther away from equilibrium. This means the deviation magnitude is increasing. The strategy interprets this as reflexive / self-reinforcing behaviour.
- if $R_t < 0$, the market is moving back toward equilibrium. This means the deviation magnitude is decreasing. The strategy interprets this as trend decay or reversion.

Remark. In continuous time, if $D(t)$ denotes the deviation process, then

$$\frac{d}{dt}\left(\frac{1}{2}D(t)^2\right) = D(t)\frac{dD}{dt}$$

Hence, R_t is the discrete analogue of the derivative of $\frac{1}{2}D(t)^2$, which can be viewed as the "energy" of deviation.

2.4 Interpretation of the four cases

The sign combinations of D_t and V_t give four economically meaningful cases:

- $D_t > 0$ and $V_t > 0$: price is above equilibrium and moving even farther above it. This is bullish self-reinforcement.
- $D_t > 0$ and $V_t < 0$: price is above equilibrium but moving back toward it. This is bullish trend decay.
- $D_t < 0$ and $V_t < 0$: price is below equilibrium and moving even farther below it. This is bearish self-reinforcement.
- $D_t < 0$ and $V_t > 0$: price is below equilibrium but recovering toward equilibrium. This is bearish trend decay.

Therefore, the sign of $R_t = D_t V_t$ is a compact way to distinguish between reinforcement and decay.

3 Signal logic

The strategy uses a threshold $\theta > 0$ to filter out small noisy moves. The trading logic is:

- if $R_t > \theta$ and $D_t > 0$, the strategy buys. This means price is above equilibrium and the positive deviation is still expanding.
- if $R_t > \theta$ and $D_t < 0$, the strategy sells / exits. This means price is below equilibrium and the negative deviation is still expanding.
- if $R_t < 0$, the strategy also sells / exits. This means the deviation is shrinking and the previous trend is losing strength.

In short, this alpha is not only a direction detector, but also a detector of whether the current deviation is being reinforced or being unwound.

4 Summary

The key idea of this alpha is simple:

measure the deviation from equilibrium, then check whether this deviation is growing or shrinking

A positive R_t indicates that the market is moving farther away from equilibrium in the same direction as the current displacement, which is interpreted as a reflexive trend. A negative R_t indicates that the market is moving back toward equilibrium, which is interpreted as trend decay or reversion.